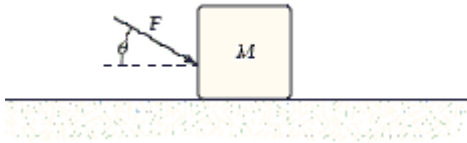


MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

- 1) A block is pushed across a horizontal surface by the force shown. If the coefficient of kinetic friction between the block and the surface is 0.30, $F = 20 \text{ N}$, $\theta = 30^\circ$, and $M = 3.0 \text{ kg}$, what is the magnitude of the acceleration of the block? 1) _____



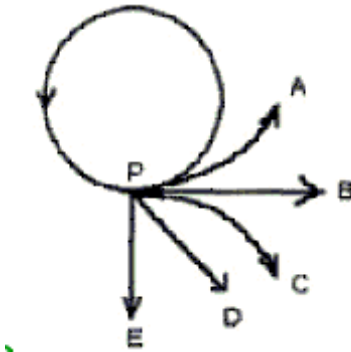
- A) 2.8 m/s^2 B) 5.4 m/s^2 E) 3.3 m/s^2
C) 2.3 m/s^2 D) 1.8 m/s^2



- 2) What is F? 2) _____
A) 100N B) 10N C) 50N D) 25N

- 3) your mass is 60kg (weight is 600N). You are standing on a bathroom scale reading kg place on a slope that makes 30 degrees with the horizontal. you are reading 3) _____
A) 60kg B) 52kg C) 30kg D) 25kg

- 4) BA girl attaches a rock to a string, which she then swings counter-clockwise in a horizontal circle. The string breaks at point P in the figure, which shows a bird's-eye view (as seen from above). Which path (A-E) will the rock follow? 4) _____



- A) path D B) path E E) path C
C) path B D) path A

- 5) A karate expert executes a swift blow and breaks a cement block with her bare hand. The magnitude of the force on her hand is 5) _____
A) the same as the force applied to the block.
B) zero.
C) more than the force applied to the block.
D) less than the force applied to the cement block.
E) need more information

- 6) The horizontal surface on which the objects slide is frictionless. If $M = 1.0 \text{ kg}$ and the magnitude of the force of the small block on the large block is 5.2 N , determine F .

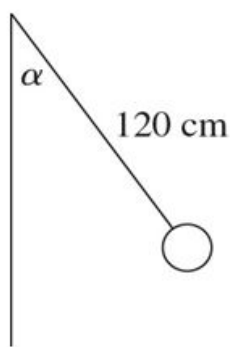
6) _____



- A) 4.8 N E) 4.1 N B) 6 N C) 7.8 N D) 9 N

- 7) A 20-g bead is attached to a light 120-cm -long string as shown in the figure. This bead moves in a horizontal circle with a constant speed of 1.5 m/s . What is the tension force that the string exerts on the bead if the angle α is measured to be 25° ?

7) _____



- A) 0.46 N B) 0.041 N C) 0.089 N D) 0.20 N E) 0.22 N

- 8) A push on a 1-kg brick accelerates it. Neglecting friction, equally accelerating a 10-kg brick requires
 A) 10 times as much force. B) one-tenth the amount of force.
 C) 100 times as much force. D) just as much force.

8) _____

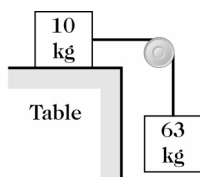
- 9) An 8.0-kg object rests on the floor of an elevator which is accelerating downward at a rate of 1.3 m/s^2 . What is the magnitude of the force the object exerts on the floor of the elevator?

9) _____

- A) 59 N E) 89 N B) 68 N
 C) 78 N D) 10 N

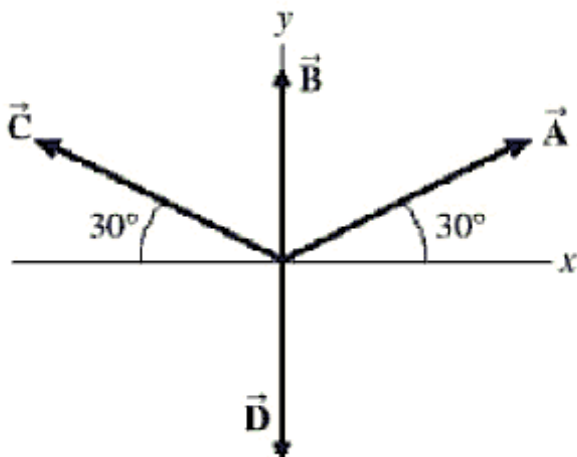
- 10) As shown in the figure, a 10-kg block on a perfectly smooth horizontal table is connected by a horizontal string to a 63-kg block that is hanging over the edge of the table. What is the magnitude of the acceleration of the 10-kg block when the other block is gently released?

10) _____

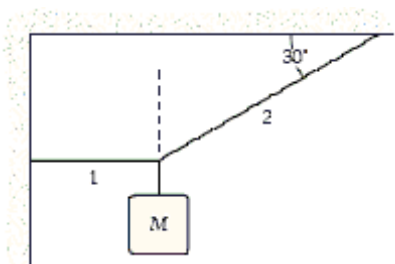


- A) 9.0 m/s^2 B) 8.5 m/s^2 C) 8.1 m/s^2 D) 7.5 m/s^2

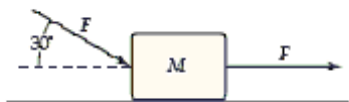
- 11) The figure below shows four vectors **A**, **B**, **C**, **D** and , having magnitudes 12.0 m, 10.0 m, 8.0 m, and 4.0 m, respectively. The sum of these four vectors is 11) _____



- A) 19.5 m at an angle 12.3° with respect to +x-axis.
 B) 19.5 m at an angle 77.8° with respect to +x-axis
 C) 16.4 m at an angle 12.3° with respect to +x-axis. E) 8.20 m at an angle 77.8° with respect to +x-axis
 D) 16.4 m at an angle 77.8° with respect to +x-axis
- 12) A satellite is in orbit around the earth. Which one feels the greater force? 12) _____
 A) the earth because the satellite has so little mass
 B) It depends on the distance of the satellite from Earth.
 C) Earth and the satellite feel exactly the same force.
 D) the satellite because the earth is so much more massive
- 13) If $M = 2.0$ kg, what is the tension in string 1? 13) _____



- A) 11N B) 34N C) 3.5N D) 40N E) 1.2N
- 14) The horizontal surface on which the block slides is frictionless. If $F = 20$ N and $M = 5.0$ kg, what is the magnitude of the resulting acceleration of the block? 14) _____

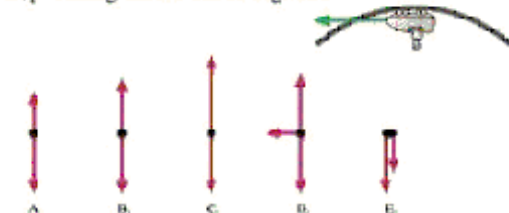


- A) 7.5m/s/s B) 5.3m/s/s
 C) 4.7m/s/s D) 6.2m/s/s
 E) 3.2m/s/s

- 15) A child is sitting on the outer edge of a merry-go-round that is 18 m in diameter. If the merry-go-round makes 8.3 rev/min, what is the velocity of the child in m/s? 15) _____
- A) 1.2 m/s B) 15.6 m/s C) 7.8 m/s D) 5.5 m/s

- 16) A small crate (m) is racing with a large crate (M) on the same frictionless slope. They slide down the slope and their starting point is the same. $M > m$. What is true? 16) _____
- A) m has the larger acceleration B) It depends on the angle of the slope
C) both crates have the same acceleration D) M has a larger acceleration

- 17) A 100-kg box is placed on a ramp. As one end of the ramp is raised, the box begins to move downward just as the angle of inclination reaches 15° . What is the coefficient of static friction between box and ramp? 17) _____
- ($g = 9.80 \text{ m/s}^2$)
A) 0.15 B) 0.27 C) 0.95 D) 0.77

- A roller coaster car does a loop-the-loop. Which of the free-body diagrams shows the forces on the car at the top of the loop? Rolling friction can be neglected.
- 
- 18) _____
- A) E B) B C) C D) D
E) A

- 19) When a falling object has reached its terminal velocity, its acceleration is 19) _____
- A) 0 B) constant C) decreasing D) g

- 20) The force required to maintain a constant velocity for an astronaut in free space is equal to 20) _____
- A) the mass of the astronaut. B) 0
C) the weight of the astronaut. D) the force required to stop the astronaut.

Answer Key

Testname: TEST2_FALL24

- 1) D
- 2) D
- 3) B
- 4) C
- 5) A
- 6) C
- 7) E
- 8) A
- 9) B
- 10) B
- 11) D
- 12) C
- 13) B
- 14) A
- 15) C
- 16) C
- 17) B
- 18) A
- 19) A
- 20) B